TOPS Technologies

Interview Questions — Data Visualization with Matplotlib

Total Questions: 46

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1. What is data visualization, and why is it important in data analytics?

> **Hint:** Think about how visuals help make sense of large or complex data.

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2. When would you choose to use Matplotlib over Seaborn for creating a chart?

> **Hint:** Focus on flexibility, customization, and level of control.

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3. Explain the difference between a Figure, Axes, and Axis in Matplotlib.

> **Hint:** Each term refers to a different part of the plotting structure.

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4. Describe the basic workflow for creating a plot using Matplotlib.

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5. How do you create a simple line chart in Matplotlib using `plt.plot()`?

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6. Which parameters of `plt.plot()` would you adjust to change the color, marker style, and line width of a line?

> **Hint:** Look for color, marker, and linewidth arguments.

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7. How would you add a title and axis labels to a Matplotlib chart?

> **Hint:** There are specific functions to set these elements.

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8. Write a code snippet to plot monthly sales data as a line chart, with circle markers and a red line.

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9. What is the difference between a vertical bar chart and a horizontal bar chart in Matplotlib?

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10. How do you create a horizontal bar chart using Matplotlib?

> **Hint:** There is a specific function for horizontal bars.

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11. What parameter would you use to change the width of the bars in a bar chart?

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12. How can you add text labels to each bar in a bar chart?

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13. Write a code snippet to plot regional revenue data as a vertical bar chart, and add value labels on top of each bar.

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14. What is the purpose of a pie chart, and when is it most appropriate to use one?

> **Hint:** Think about showing parts of a whole.

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15. How do you display percentage values on each segment in a pie chart using Matplotlib?

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16. How can you highlight or 'explode' one segment of a pie chart?

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17. Describe how to create a donut chart in Matplotlib.

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18. Write a code snippet to visualize market share data as a pie chart with exploded largest segment and percentage labels.

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19. What does a histogram show, and how is it different from a bar chart?

> **Hint:** Think about data distribution vs categorical values.

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20. How do you control the number of bins in a histogram in Matplotlib?

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21. Explain the difference between plotting density and frequency in a histogram.

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22. Write code to plot a histogram of customer ages with 10 bins.

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23. What kind of data relationship does a scatter plot show?

> **Hint:** Think about how two numeric variables relate.

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24. How can you add a trend line to a scatter plot in Matplotlib?

> **Hint:** Consider fitting a line using numpy.

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25. Write code to plot advertising spend vs sales as a scatter plot and add a regression line.

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26. What is the difference between `plt.subplot()` and `plt.subplots()`?

> **Hint:** One returns axes objects, the other sets up the grid.

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27. How can you create a 2x2 grid of plots with shared x-axis using Matplotlib?

> **Hint:** Look for parameters in plt.subplots().

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28. Write code to display four different charts in a single figure using subplots.

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29. How do you change the overall figure size in a Matplotlib plot?

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30. What is the purpose of `plt.style.use()` in Matplotlib?

> **Hint:** Think about themes and appearance.

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31. How can you customize the color palette of your charts in Matplotlib?

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32. Write code to plot a sales trend chart using the 'ggplot' style and a custom color.

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33. Explain what a stacked bar chart is and when you might use one.

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34. How can you add error bars to a line or bar chart in Matplotlib?

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35. Describe how to plot a time series chart in Matplotlib.

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36. Write code to create an area chart showing monthly sales data.

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37. How do you add annotations (such as arrows or text boxes) to highlight specific points in a chart?

> **Hint:** There is a function specifically for annotations.

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38. How can you position the legend in a Matplotlib chart?

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39. Write code to annotate the highest sales month in a line chart with an arrow.

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40. What function is used to save a Matplotlib plot to a file, and how do you specify the file format?

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41. How can you control the resolution (DPI) of a saved plot in Matplotlib?

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42. Describe how you would export a chart from Matplotlib to embed it in a PowerPoint presentation.

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43. Given a retail sales dataset, which types of charts would you use to show KPIs like total sales, sales by category, and customer age distribution?

> **Hint:** Think about trend, category, and distribution visuals.

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44. Explain how you would integrate annotations and export options when preparing a visual analytics report using Matplotlib.

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45. How could you use an AI tool like ChatGPT or Copilot to help you write or debug Matplotlib code, and what steps would you take to verify the code works as expected?

> **Hint:** Think about code suggestions and checking output or documentation.

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46. In what ways could generative AI tools speed up your data visualization workflow, and where would you still rely on your own skills and judgment?
